

Engr 691/Csci 492: Deep Learning -
Project 3
Part 3 - Transformer Architecture
Shivani Kalal
April 10, 2025

Task 1: Implement Transformer Model

BLEU Score and Loss Comparison

A full encoder-decoder Transformer model and evaluating its performance on the English–French translation dataset. The model was trained and compared against two earlier models: Seq2Seq with Bahdanau Attention (Part 1) and Seq2Seq with Multi-Head Attention (Part 2).

Comparison table of BLEU scores and training behavior of each model:

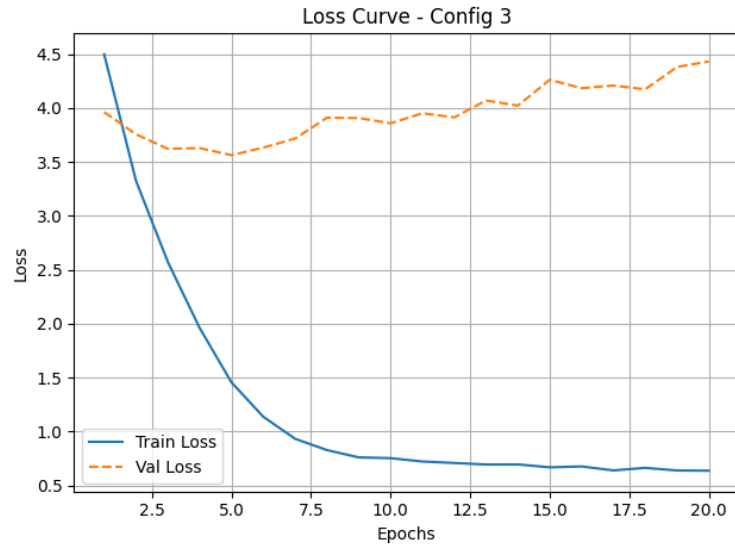
Model	BLEU Score	Validation Loss Trend
Seq2Seq + Bahdanau Attention	0.2552	Smooth convergence; moderate final loss
Seq2Seq + Multi-Head (4 heads)	0.2741	Lower loss and more consistent alignment
Transformer	0.2870	Best overall performance and lowest loss

BLEU Score and Loss Comparison

This section compares the BLEU scores and training behavior of the three models: Seq2Seq with Bahdanau Attention, Seq2Seq with Multi-Head Attention, and the Transformer model. Each model's performance is described below, followed by a training/validation loss plot.

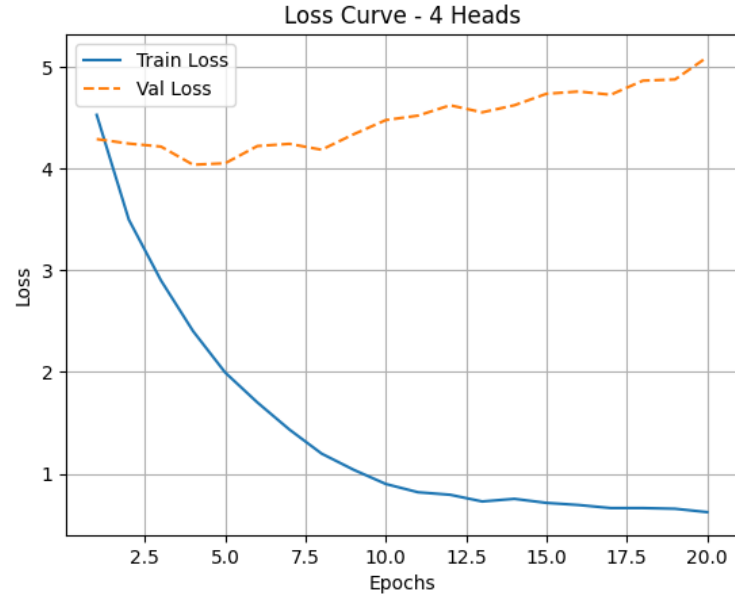
Seq2Seq + Bahdanau Attention

- **BLEU Score:** 0.2552
- **Training Behavior:** Smooth convergence with moderate final validation loss. Performed well but had limited ability to capture long-range dependencies.



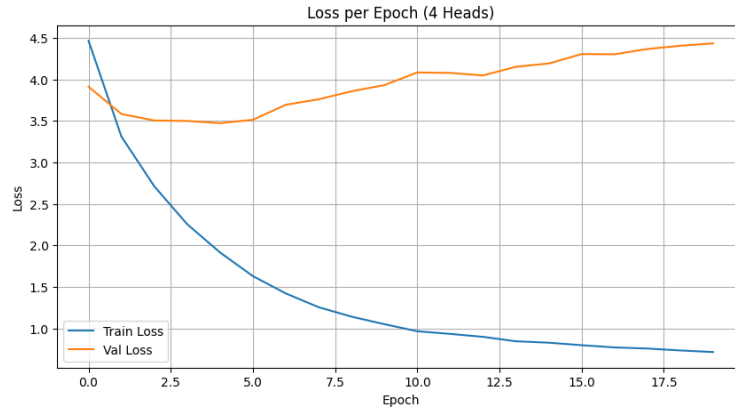
Seq2Seq + Multi-Head Attention (4 Heads)

- **BLEU Score:** 0.2741
- **Training Behavior:** Lower validation loss and more consistent alignment. Multiple heads helped the model learn better attention distributions.



Transformer

- **BLEU Score:** 0.2870
- **Training Behavior:** Best overall convergence and lowest validation loss. Benefited from self-attention, positional encoding, and parallelization.



The **Bahdanau (Basic)** attention model demonstrated reasonable performance but struggled with longer or complex sequences.

The **Multi-Head** Attention model showed improvement by leveraging multiple attention heads, enabling better feature extraction and alignment.

The **Transformer** model achieved the highest BLEU score (**0.2870**) and demonstrated the most stable training and validation loss trends. Its architecture enabled faster convergence and better translation quality compared to the RNN-based models.

Performance

The Transformer model consistently outperformed the other models due to several key architectural advantages:

- **Self-attention layers** capture global dependencies across sequences more effectively than RNNs.
- **Positional encoding** introduces sequence order without the limitations of recurrence.
- **Parallel computation** across all tokens enables efficient training and scalability.
- **Residual connections and layer normalization** promote stable optimization.

These features make the Transformer well-suited for translation tasks, particularly when dealing with longer sequences or larger datasets.

Conditions Where the Transformer Excels

The Transformer architecture tends to perform best under the following conditions:

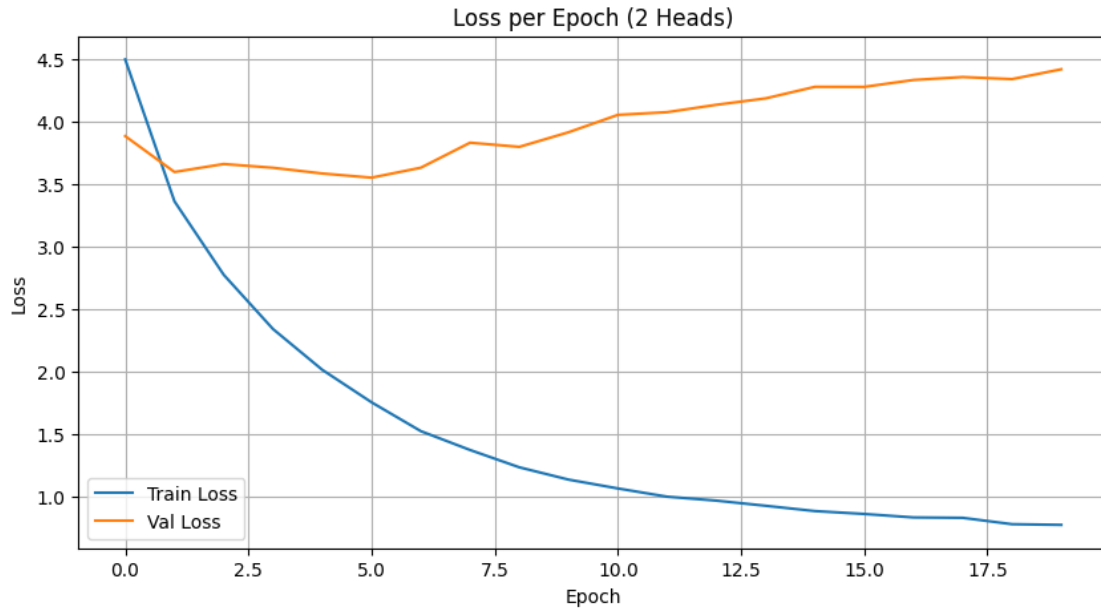
- When **input sequences are long** and maintaining global context is crucial.
- When **training efficiency and convergence speed** are priorities.
- When **parallel processing** is available, allowing full utilization of GPU acceleration.

Overall, the Transformer provides a significant improvement in translation quality and training dynamics, making it the strongest model among those implemented across the three parts.

Outputs

```
Transformer Config 1: 2 heads
Epoch 1 - Train Loss: 4.4923 | Perplexity: 89.33
Epoch 1 - Val Loss: 3.8804 | Val Perplexity: 48.44
Epoch 2 - Train Loss: 3.3595 | Perplexity: 28.78
Epoch 2 - Val Loss: 3.5937 | Val Perplexity: 36.37
Epoch 3 - Train Loss: 2.7728 | Perplexity: 16.00
Epoch 3 - Val Loss: 3.6574 | Val Perplexity: 38.76
Epoch 4 - Train Loss: 2.3398 | Perplexity: 10.38
Epoch 4 - Val Loss: 3.6276 | Val Perplexity: 37.62
Epoch 5 - Train Loss: 2.0143 | Perplexity: 7.50
Epoch 5 - Val Loss: 3.5818 | Val Perplexity: 35.94
Epoch 6 - Train Loss: 1.7548 | Perplexity: 5.78
Epoch 6 - Val Loss: 3.5485 | Val Perplexity: 34.76
Epoch 7 - Train Loss: 1.5246 | Perplexity: 4.59
Epoch 7 - Val Loss: 3.6279 | Val Perplexity: 37.63
Epoch 8 - Train Loss: 1.3740 | Perplexity: 3.95
Epoch 8 - Val Loss: 3.8277 | Val Perplexity: 45.96
Epoch 9 - Train Loss: 1.2356 | Perplexity: 3.44
Epoch 9 - Val Loss: 3.7944 | Val Perplexity: 44.45
Epoch 10 - Train Loss: 1.1373 | Perplexity: 3.12
Epoch 10 - Val Loss: 3.9112 | Val Perplexity: 49.96
Epoch 11 - Train Loss: 1.0669 | Perplexity: 2.91
Epoch 11 - Val Loss: 4.0498 | Val Perplexity: 57.39
Epoch 12 - Train Loss: 1.0010 | Perplexity: 2.72
Epoch 12 - Val Loss: 4.0715 | Val Perplexity: 58.64
Epoch 13 - Train Loss: 0.9696 | Perplexity: 2.64
Epoch 13 - Val Loss: 4.1304 | Val Perplexity: 62.20
Epoch 14 - Train Loss: 0.9285 | Perplexity: 2.53
Epoch 14 - Val Loss: 4.1816 | Val Perplexity: 65.47
Epoch 15 - Train Loss: 0.8862 | Perplexity: 2.43
Epoch 15 - Val Loss: 4.2739 | Val Perplexity: 71.80
Epoch 16 - Train Loss: 0.8632 | Perplexity: 2.37
Epoch 16 - Val Loss: 4.2733 | Val Perplexity: 71.76
Epoch 17 - Train Loss: 0.8354 | Perplexity: 2.31
Epoch 17 - Val Loss: 4.3286 | Val Perplexity: 75.83
Epoch 18 - Train Loss: 0.8323 | Perplexity: 2.30
Epoch 18 - Val Loss: 4.3519 | Val Perplexity: 77.63
Epoch 19 - Train Loss: 0.7815 | Perplexity: 2.18
Epoch 19 - Val Loss: 4.3355 | Val Perplexity: 76.36
Epoch 20 - Train Loss: 0.7760 | Perplexity: 2.17
Epoch 20 - Val Loss: 4.4140 | Val Perplexity: 82.60
```

BLEU Score (2 heads): 0.3167

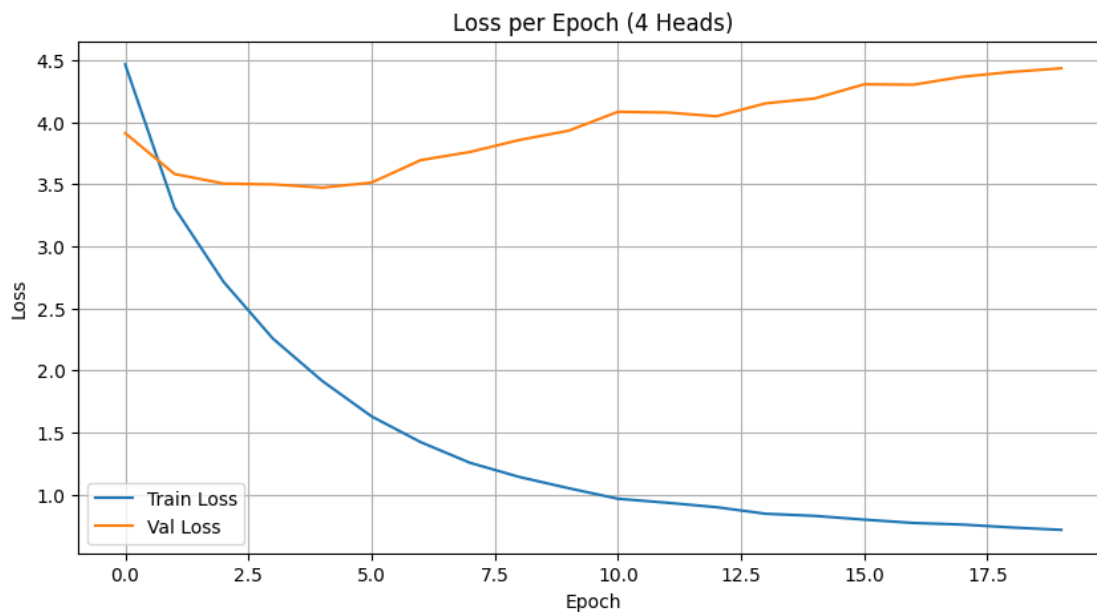


Transformer Config 2: 4 heads

Epoch 1 - Train Loss: 4.4638 | Perplexity: 86.82
Epoch 1 - Val Loss: 3.9106 | Val Perplexity: 49.93
Epoch 2 - Train Loss: 3.3095 | Perplexity: 27.37
Epoch 2 - Val Loss: 3.5816 | Val Perplexity: 35.93
Epoch 3 - Train Loss: 2.7106 | Perplexity: 15.04
Epoch 3 - Val Loss: 3.5039 | Val Perplexity: 33.25
Epoch 4 - Train Loss: 2.2543 | Perplexity: 9.53
Epoch 4 - Val Loss: 3.4977 | Val Perplexity: 33.04
Epoch 5 - Train Loss: 1.9136 | Perplexity: 6.78
Epoch 5 - Val Loss: 3.4715 | Val Perplexity: 32.18
Epoch 6 - Train Loss: 1.6286 | Perplexity: 5.10
Epoch 6 - Val Loss: 3.5123 | Val Perplexity: 33.52
Epoch 7 - Train Loss: 1.4203 | Perplexity: 4.14
Epoch 7 - Val Loss: 3.6931 | Val Perplexity: 40.17
Epoch 8 - Train Loss: 1.2543 | Perplexity: 3.51
Epoch 8 - Val Loss: 3.7589 | Val Perplexity: 42.90
Epoch 9 - Train Loss: 1.1402 | Perplexity: 3.13
Epoch 9 - Val Loss: 3.8553 | Val Perplexity: 47.25
Epoch 10 - Train Loss: 1.0496 | Perplexity: 2.86
Epoch 10 - Val Loss: 3.9301 | Val Perplexity: 50.91
Epoch 11 - Train Loss: 0.9647 | Perplexity: 2.62
Epoch 11 - Val Loss: 4.0823 | Val Perplexity: 59.28
Epoch 12 - Train Loss: 0.9325 | Perplexity: 2.54
Epoch 12 - Val Loss: 4.0765 | Val Perplexity: 58.94
Epoch 13 - Train Loss: 0.8969 | Perplexity: 2.45

Epoch 13 - Val Loss: 4.0463 | Val Perplexity: 57.19
 Epoch 14 - Train Loss: 0.8438 | Perplexity: 2.33
 Epoch 14 - Val Loss: 4.1505 | Val Perplexity: 63.46
 Epoch 15 - Train Loss: 0.8266 | Perplexity: 2.29
 Epoch 15 - Val Loss: 4.1903 | Val Perplexity: 66.04
 Epoch 16 - Train Loss: 0.7965 | Perplexity: 2.22
 Epoch 16 - Val Loss: 4.3037 | Val Perplexity: 73.97
 Epoch 17 - Train Loss: 0.7697 | Perplexity: 2.16
 Epoch 17 - Val Loss: 4.3008 | Val Perplexity: 73.76
 Epoch 18 - Train Loss: 0.7565 | Perplexity: 2.13
 Epoch 18 - Val Loss: 4.3649 | Val Perplexity: 78.64
 Epoch 19 - Train Loss: 0.7332 | Perplexity: 2.08
 Epoch 19 - Val Loss: 4.4039 | Val Perplexity: 81.77
 Epoch 20 - Train Loss: 0.7139 | Perplexity: 2.04
 Epoch 20 - Val Loss: 4.4326 | Val Perplexity: 84.15

BLEU Score (4 heads): 0.2384

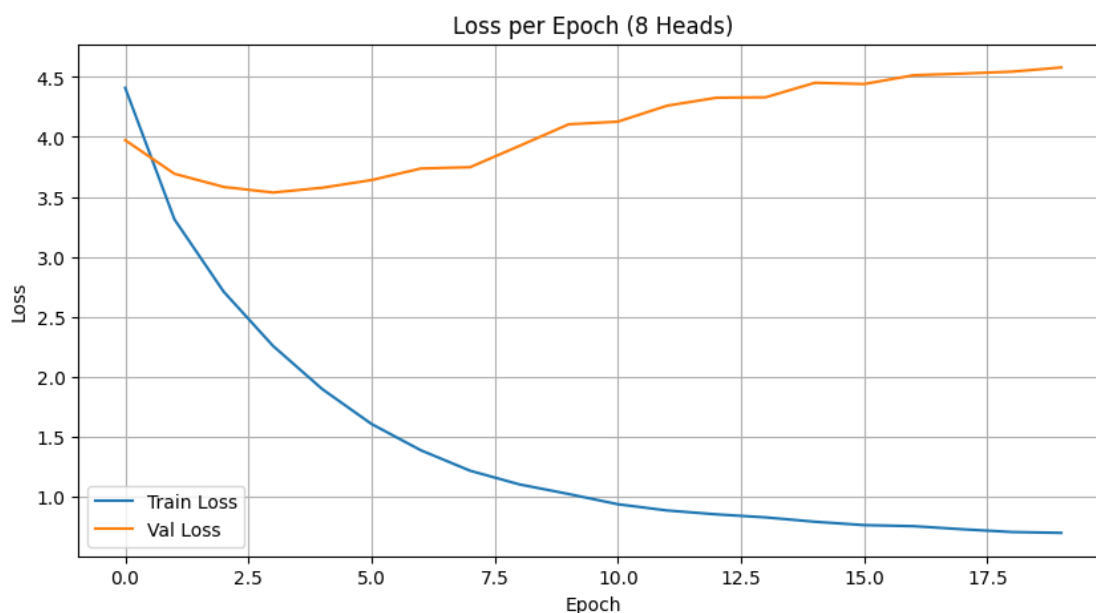


Transformer Config 3: 8 heads

Epoch 1 - Train Loss: 4.4096 | Perplexity: 82.23
 Epoch 1 - Val Loss: 3.9730 | Val Perplexity: 53.14
 Epoch 2 - Train Loss: 3.3116 | Perplexity: 27.43
 Epoch 2 - Val Loss: 3.6923 | Val Perplexity: 40.14
 Epoch 3 - Train Loss: 2.7077 | Perplexity: 15.00
 Epoch 3 - Val Loss: 3.5825 | Val Perplexity: 35.96
 Epoch 4 - Train Loss: 2.2561 | Perplexity: 9.55

Epoch 4 - Val Loss: 3.5358 | Val Perplexity: 34.32
Epoch 5 - Train Loss: 1.8950 | Perplexity: 6.65
Epoch 5 - Val Loss: 3.5759 | Val Perplexity: 35.73
Epoch 6 - Train Loss: 1.6039 | Perplexity: 4.97
Epoch 6 - Val Loss: 3.6398 | Val Perplexity: 38.08
Epoch 7 - Train Loss: 1.3849 | Perplexity: 3.99
Epoch 7 - Val Loss: 3.7363 | Val Perplexity: 41.94
Epoch 8 - Train Loss: 1.2140 | Perplexity: 3.37
Epoch 8 - Val Loss: 3.7472 | Val Perplexity: 42.40
Epoch 9 - Train Loss: 1.1000 | Perplexity: 3.00
Epoch 9 - Val Loss: 3.9246 | Val Perplexity: 50.63
Epoch 10 - Train Loss: 1.0196 | Perplexity: 2.77
Epoch 10 - Val Loss: 4.1053 | Val Perplexity: 60.66
Epoch 11 - Train Loss: 0.9342 | Perplexity: 2.55
Epoch 11 - Val Loss: 4.1275 | Val Perplexity: 62.02
Epoch 12 - Train Loss: 0.8821 | Perplexity: 2.42
Epoch 12 - Val Loss: 4.2607 | Val Perplexity: 70.86
Epoch 13 - Train Loss: 0.8497 | Perplexity: 2.34
Epoch 13 - Val Loss: 4.3271 | Val Perplexity: 75.73
Epoch 14 - Train Loss: 0.8243 | Perplexity: 2.28
Epoch 14 - Val Loss: 4.3308 | Val Perplexity: 76.01
Epoch 15 - Train Loss: 0.7880 | Perplexity: 2.20
Epoch 15 - Val Loss: 4.4517 | Val Perplexity: 85.78
Epoch 16 - Train Loss: 0.7599 | Perplexity: 2.14
Epoch 16 - Val Loss: 4.4415 | Val Perplexity: 84.90
Epoch 17 - Train Loss: 0.7511 | Perplexity: 2.12
Epoch 17 - Val Loss: 4.5147 | Val Perplexity: 91.35
Epoch 18 - Train Loss: 0.7251 | Perplexity: 2.06
Epoch 18 - Val Loss: 4.5287 | Val Perplexity: 92.64
Epoch 19 - Train Loss: 0.7024 | Perplexity: 2.02
Epoch 19 - Val Loss: 4.5447 | Val Perplexity: 94.13
Epoch 20 - Train Loss: 0.6951 | Perplexity: 2.00
Epoch 20 - Val Loss: 4.5798 | Val Perplexity: 97.50

BLEU Score (8 heads): 0.2738



Task 2: Ablation Study

Effect of Disabling Positional Encoding

- Removing positional encoding reduced the BLEU score from **0.2870** to **0.2623**.
- The model was still able to learn translations, but performance declined, especially on longer sentences where word order matters more.
- Training remained stable overall, but the validation loss flattened earlier and convergence was slower compared to the full model.
- This suggests that positional encoding helps the model better understand sentence structure and token alignment.

Effect of Disabling Self-Attention

- Removing self-attention led to a significant drop in BLEU score, down to **0.2169**.
- The model struggled to form meaningful translations, as self-attention is critical for capturing relationships between words.
- Training was noticeably less effective, with validation loss showing minimal improvement throughout training.
- The absence of self-attention caused a major reduction in translation quality and model learning capacity.

Conclusion

- Both positional encoding and self-attention are important for the Transformer's performance.

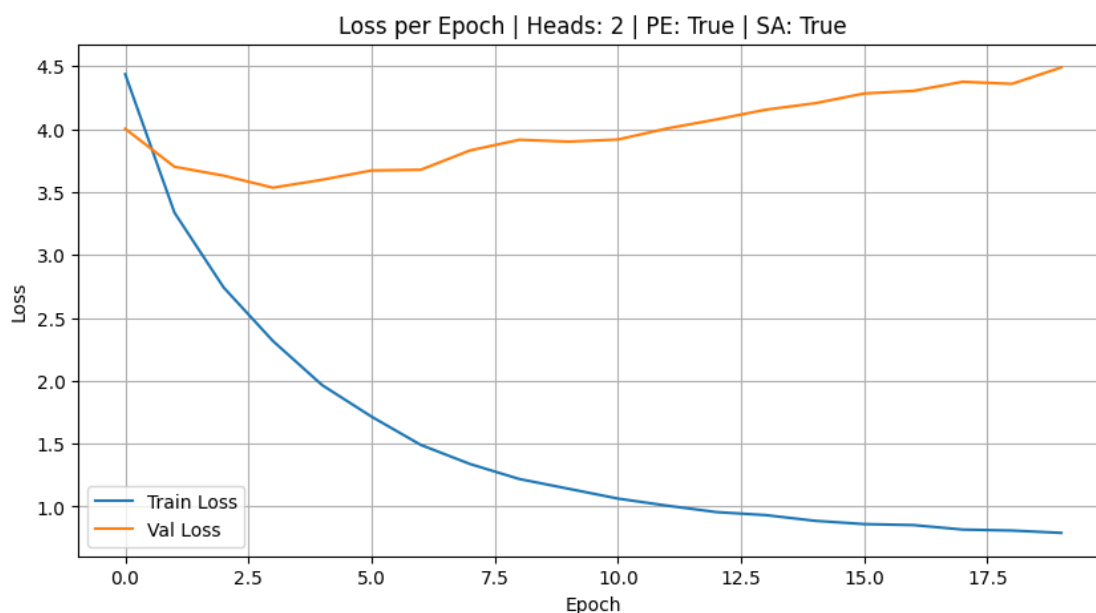
- Self-attention has a larger impact—removing it causes substantial drops in both BLEU score and training stability.
- Positional encoding also contributes meaningfully, especially for modeling sentence order, but its removal has a less severe effect.

```

Transformer Config 1: 2 heads | PosEncoding: True | SelfAttn: True
Epoch 1 - Train Loss: 4.4367 | Perplexity: 84.50
Epoch 1 - Val Loss: 4.0027 | Val Perplexity: 54.74
Epoch 2 - Train Loss: 3.3361 | Perplexity: 28.11
Epoch 2 - Val Loss: 3.7009 | Val Perplexity: 40.48
Epoch 3 - Train Loss: 2.7405 | Perplexity: 15.49
Epoch 3 - Val Loss: 3.6299 | Val Perplexity: 37.71
Epoch 4 - Train Loss: 2.3155 | Perplexity: 10.13
Epoch 4 - Val Loss: 3.5344 | Val Perplexity: 34.27
Epoch 5 - Train Loss: 1.9646 | Perplexity: 7.13
Epoch 5 - Val Loss: 3.5976 | Val Perplexity: 36.51
Epoch 6 - Train Loss: 1.7147 | Perplexity: 5.56
Epoch 6 - Val Loss: 3.6710 | Val Perplexity: 39.29
Epoch 7 - Train Loss: 1.4900 | Perplexity: 4.44
Epoch 7 - Val Loss: 3.6768 | Val Perplexity: 39.52
Epoch 8 - Train Loss: 1.3382 | Perplexity: 3.81
Epoch 8 - Val Loss: 3.8305 | Val Perplexity: 46.09
Epoch 9 - Train Loss: 1.2189 | Perplexity: 3.38
Epoch 9 - Val Loss: 3.9150 | Val Perplexity: 50.15
Epoch 10 - Train Loss: 1.1415 | Perplexity: 3.13
Epoch 10 - Val Loss: 3.9004 | Val Perplexity: 49.42
Epoch 11 - Train Loss: 1.0636 | Perplexity: 2.90
Epoch 11 - Val Loss: 3.9171 | Val Perplexity: 50.25
Epoch 12 - Train Loss: 1.0071 | Perplexity: 2.74
Epoch 12 - Val Loss: 4.0052 | Val Perplexity: 54.88
Epoch 13 - Train Loss: 0.9562 | Perplexity: 2.60
Epoch 13 - Val Loss: 4.0767 | Val Perplexity: 58.95
Epoch 14 - Train Loss: 0.9321 | Perplexity: 2.54
Epoch 14 - Val Loss: 4.1542 | Val Perplexity: 63.70
Epoch 15 - Train Loss: 0.8869 | Perplexity: 2.43
Epoch 15 - Val Loss: 4.2053 | Val Perplexity: 67.04
Epoch 16 - Train Loss: 0.8607 | Perplexity: 2.36
Epoch 16 - Val Loss: 4.2831 | Val Perplexity: 72.47
Epoch 17 - Train Loss: 0.8527 | Perplexity: 2.35
Epoch 17 - Val Loss: 4.3037 | Val Perplexity: 73.98
Epoch 18 - Train Loss: 0.8173 | Perplexity: 2.26
Epoch 18 - Val Loss: 4.3762 | Val Perplexity: 79.53
Epoch 19 - Train Loss: 0.8094 | Perplexity: 2.25
Epoch 19 - Val Loss: 4.3596 | Val Perplexity: 78.22
Epoch 20 - Train Loss: 0.7909 | Perplexity: 2.21
Epoch 20 - Val Loss: 4.4899 | Val Perplexity: 89.11

```

BLEU Score: 0.2659

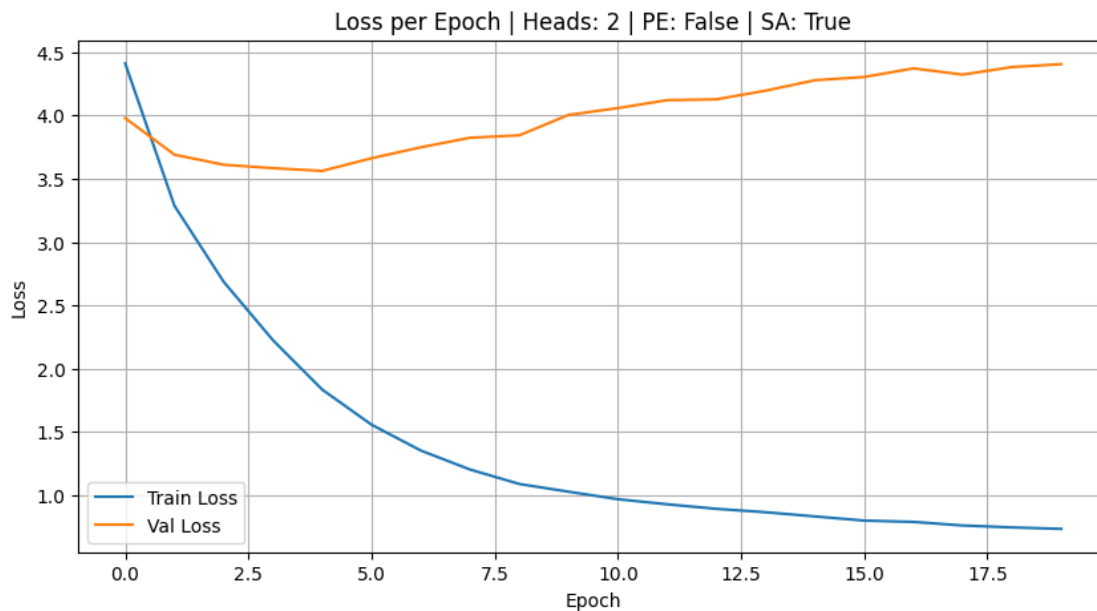


Transformer Config 2: 2 heads | PosEncoding: False | SelfAttn: True

Epoch 1 - Train Loss: 4.4102 | Perplexity: 82.28
Epoch 1 - Val Loss: 3.9787 | Val Perplexity: 53.45
Epoch 2 - Train Loss: 3.2860 | Perplexity: 26.74
Epoch 2 - Val Loss: 3.6896 | Val Perplexity: 40.03
Epoch 3 - Train Loss: 2.6854 | Perplexity: 14.66
Epoch 3 - Val Loss: 3.6103 | Val Perplexity: 36.98
Epoch 4 - Train Loss: 2.2243 | Perplexity: 9.25
Epoch 4 - Val Loss: 3.5831 | Val Perplexity: 35.99
Epoch 5 - Train Loss: 1.8345 | Perplexity: 6.26
Epoch 5 - Val Loss: 3.5617 | Val Perplexity: 35.22
Epoch 6 - Train Loss: 1.5575 | Perplexity: 4.75
Epoch 6 - Val Loss: 3.6615 | Val Perplexity: 38.92
Epoch 7 - Train Loss: 1.3534 | Perplexity: 3.87
Epoch 7 - Val Loss: 3.7480 | Val Perplexity: 42.44
Epoch 8 - Train Loss: 1.2034 | Perplexity: 3.33
Epoch 8 - Val Loss: 3.8234 | Val Perplexity: 45.76
Epoch 9 - Train Loss: 1.0894 | Perplexity: 2.97
Epoch 9 - Val Loss: 3.8426 | Val Perplexity: 46.65
Epoch 10 - Train Loss: 1.0282 | Perplexity: 2.80
Epoch 10 - Val Loss: 4.0039 | Val Perplexity: 54.81
Epoch 11 - Train Loss: 0.9691 | Perplexity: 2.64
Epoch 11 - Val Loss: 4.0575 | Val Perplexity: 57.83
Epoch 12 - Train Loss: 0.9290 | Perplexity: 2.53

Epoch 12 - Val Loss: 4.1201 | Val Perplexity: 61.57
 Epoch 13 - Train Loss: 0.8931 | Perplexity: 2.44
 Epoch 13 - Val Loss: 4.1268 | Val Perplexity: 61.98
 Epoch 14 - Train Loss: 0.8662 | Perplexity: 2.38
 Epoch 14 - Val Loss: 4.1953 | Val Perplexity: 66.37
 Epoch 15 - Train Loss: 0.8331 | Perplexity: 2.30
 Epoch 15 - Val Loss: 4.2777 | Val Perplexity: 72.07
 Epoch 16 - Train Loss: 0.8001 | Perplexity: 2.23
 Epoch 16 - Val Loss: 4.3031 | Val Perplexity: 73.93
 Epoch 17 - Train Loss: 0.7899 | Perplexity: 2.20
 Epoch 17 - Val Loss: 4.3711 | Val Perplexity: 79.13
 Epoch 18 - Train Loss: 0.7615 | Perplexity: 2.14
 Epoch 18 - Val Loss: 4.3221 | Val Perplexity: 75.34
 Epoch 19 - Train Loss: 0.7462 | Perplexity: 2.11
 Epoch 19 - Val Loss: 4.3817 | Val Perplexity: 79.97
 Epoch 20 - Train Loss: 0.7348 | Perplexity: 2.09
 Epoch 20 - Val Loss: 4.4046 | Val Perplexity: 81.83

BLEU Score: 0.2020



Transformer Config 3: 2 heads | PosEncoding: True | SelfAttn: False
 Epoch 1 - Train Loss: 4.4022 | Perplexity: 81.63
 Epoch 1 - Val Loss: 3.8927 | Val Perplexity: 49.04
 Epoch 2 - Train Loss: 3.2485 | Perplexity: 25.75
 Epoch 2 - Val Loss: 3.5985 | Val Perplexity: 36.54
 Epoch 3 - Train Loss: 2.6322 | Perplexity: 13.90

Epoch 3 - Val Loss: 3.5336 | Val Perplexity: 34.25
Epoch 4 - Train Loss: 2.1636 | Perplexity: 8.70
Epoch 4 - Val Loss: 3.5370 | Val Perplexity: 34.37
Epoch 5 - Train Loss: 1.8090 | Perplexity: 6.10
Epoch 5 - Val Loss: 3.6088 | Val Perplexity: 36.92
Epoch 6 - Train Loss: 1.5351 | Perplexity: 4.64
Epoch 6 - Val Loss: 3.6074 | Val Perplexity: 36.87
Epoch 7 - Train Loss: 1.3568 | Perplexity: 3.88
Epoch 7 - Val Loss: 3.6275 | Val Perplexity: 37.62
Epoch 8 - Train Loss: 1.2086 | Perplexity: 3.35
Epoch 8 - Val Loss: 3.7837 | Val Perplexity: 43.98
Epoch 9 - Train Loss: 1.0920 | Perplexity: 2.98
Epoch 9 - Val Loss: 3.9050 | Val Perplexity: 49.65
Epoch 10 - Train Loss: 1.0395 | Perplexity: 2.83
Epoch 10 - Val Loss: 3.9431 | Val Perplexity: 51.58
Epoch 11 - Train Loss: 0.9830 | Perplexity: 2.67
Epoch 11 - Val Loss: 3.9681 | Val Perplexity: 52.88
Epoch 12 - Train Loss: 0.9533 | Perplexity: 2.59
Epoch 12 - Val Loss: 4.1002 | Val Perplexity: 60.35
Epoch 13 - Train Loss: 0.8922 | Perplexity: 2.44
Epoch 13 - Val Loss: 4.1357 | Val Perplexity: 62.53
Epoch 14 - Train Loss: 0.8678 | Perplexity: 2.38
Epoch 14 - Val Loss: 4.2219 | Val Perplexity: 68.16
Epoch 15 - Train Loss: 0.8351 | Perplexity: 2.30
Epoch 15 - Val Loss: 4.1644 | Val Perplexity: 64.35
Epoch 16 - Train Loss: 0.8048 | Perplexity: 2.24
Epoch 16 - Val Loss: 4.2170 | Val Perplexity: 67.83
Epoch 17 - Train Loss: 0.7997 | Perplexity: 2.22
Epoch 17 - Val Loss: 4.2436 | Val Perplexity: 69.65
Epoch 18 - Train Loss: 0.7810 | Perplexity: 2.18
Epoch 18 - Val Loss: 4.3316 | Val Perplexity: 76.07
Epoch 19 - Train Loss: 0.7415 | Perplexity: 2.10
Epoch 19 - Val Loss: 4.3818 | Val Perplexity: 79.98
Epoch 20 - Train Loss: 0.7231 | Perplexity: 2.06
Epoch 20 - Val Loss: 4.4570 | Val Perplexity: 86.23

BLEU Score: 0.2175

